The Architecture of anatomy_segmenter.py

Step-by-Step Execution Flow in the MONAI Cardiac Fat FL Pipeline

PIPELINE ARCHITECTURE

STAGE 1: DATA PREPROCESSING

- DICOM Loading (MONAI)
- Resampling (Spacing: 1.0 mm)
- Intensity Normalization (Window/Level)
- Crop & Pad (Fixed Tensor Shape)

STAGE 2: MODEL INFERENCE (anatomy_segmenter.py)

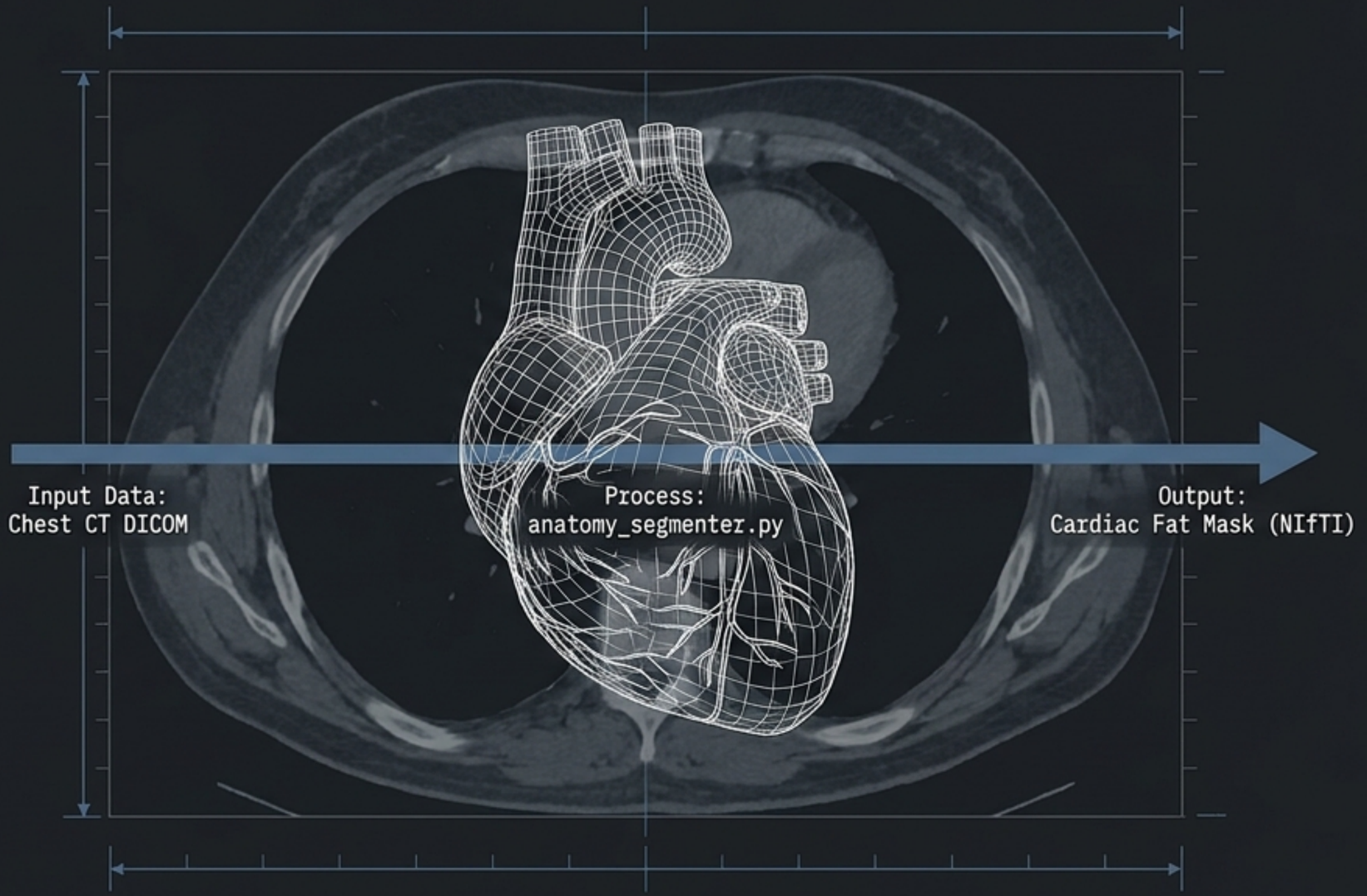
- UNet (MONAI)
  - Input Tensor: (1, 1, 128, 128, 128)
  - Activation: Softmax
- Output Channels: 2 (Background, Cardiac Fat)

STAGE 3: POST-PROCESSING

- Thresholding (>0.5)
- Morphological Operations (Opening)
- Connected Components Analysis
- Save as NIfTI (Mask)

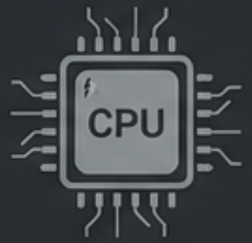
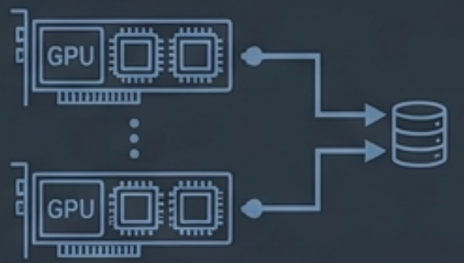
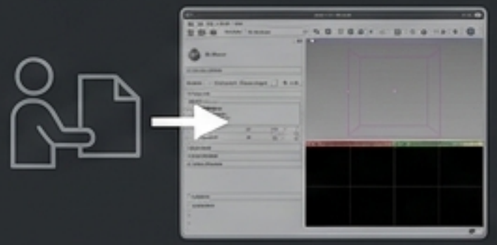
CRITICAL CONSIDERATIONS (Legacy / Bottlenecks)

- Legacy Dataloader: Slow I/O
- CPU-bound Preprocessing: High Latency
- Non-optimized Tensor Conversion



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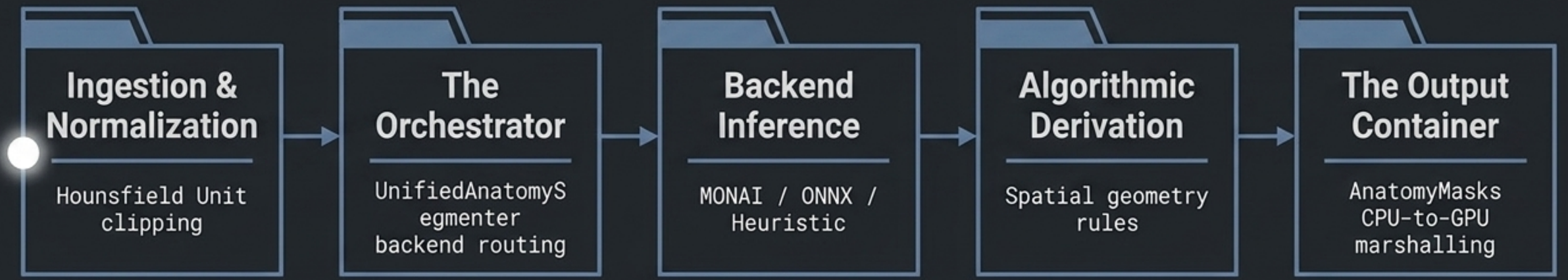
The Architectural Imperative

	Legacy System `CardiacFatQuant`	**Modernized Pipeline** `monai-cardiac-fat-fl`
Framework & Runtime	`Python 2.7 / TF 1.14 (Frozen .pb)`	`Python 3.11+ / PyTorch 2.x & ONNX GPU`
Execution Style	Single-threaded, severe CPU contention 	GPU-accelerated, safely concurrent, dynamic routing 
Integration	Manual 3D Slicer import 	Native OHIF v3+ via DICOM-SEG objects 

Architectural Goal: 100.0% exact clinical equivalence (Δ = 0.00 cm³, Δ = 0.00 HU) **with >10x throughput (120+ scans/min).**

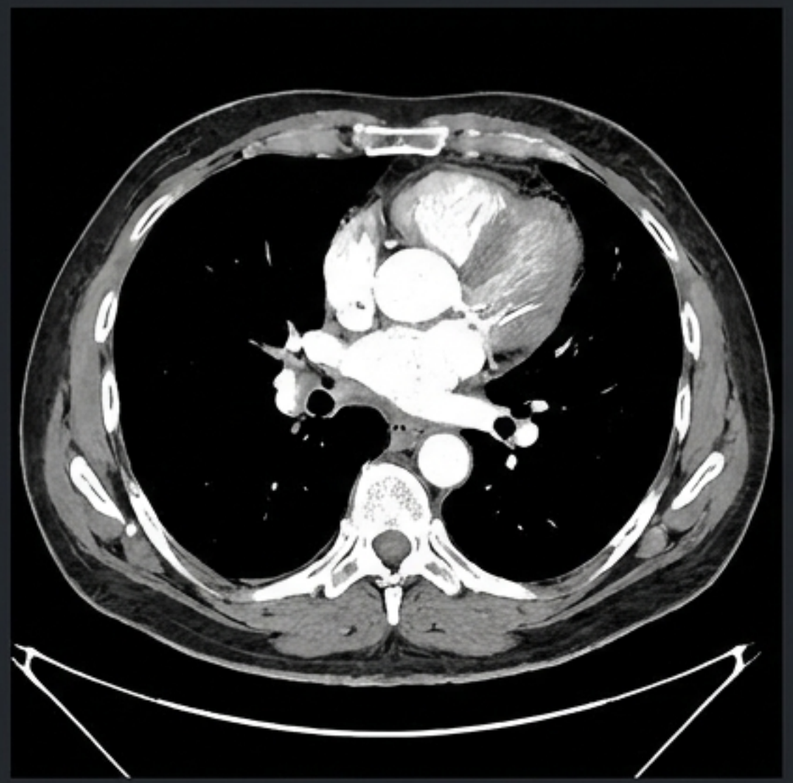
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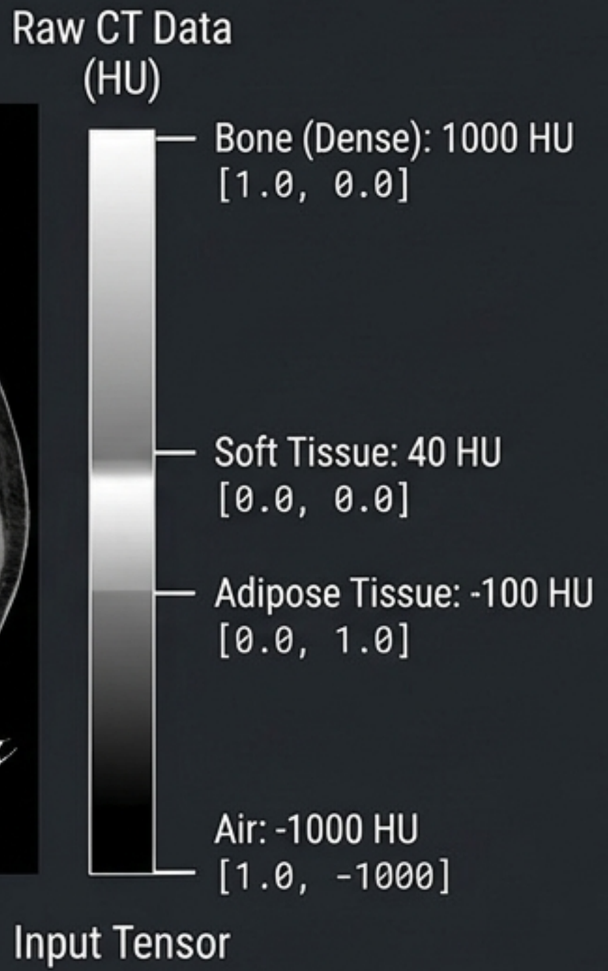
The Data Journey: High-Level Execution Flow



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Step 1: Ingestion & Hounsfield Unit Normalization





The Clinical Window

Adipose tissue is strictly bounded at `[-190.0, -30.0]` HU. The input tensor must be normalized to preserve this density fidelity.

MONAI Path Normalization

The standard cardiac window `[-200, 800]` HU is mathematically clipped and mapped to a `[0.0, 1.0]` float tensor.

ONNX Path Normalization

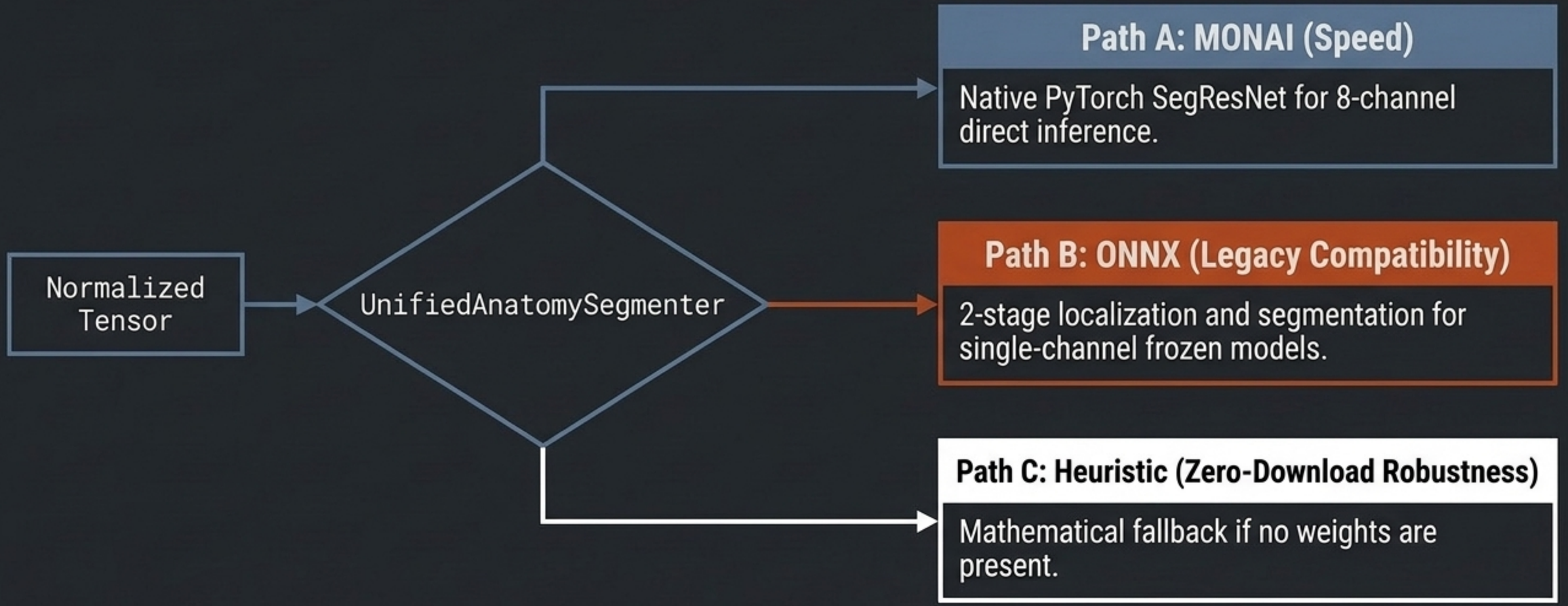
Requires a specific legacy calibration: `[-1024, 3071]` mapped via `(clip - 1023.5) / 2047.5`.

Takeaway: Strict HU adherence guarantees mathematical equivalence for the final EAT/PVAT thresholding.

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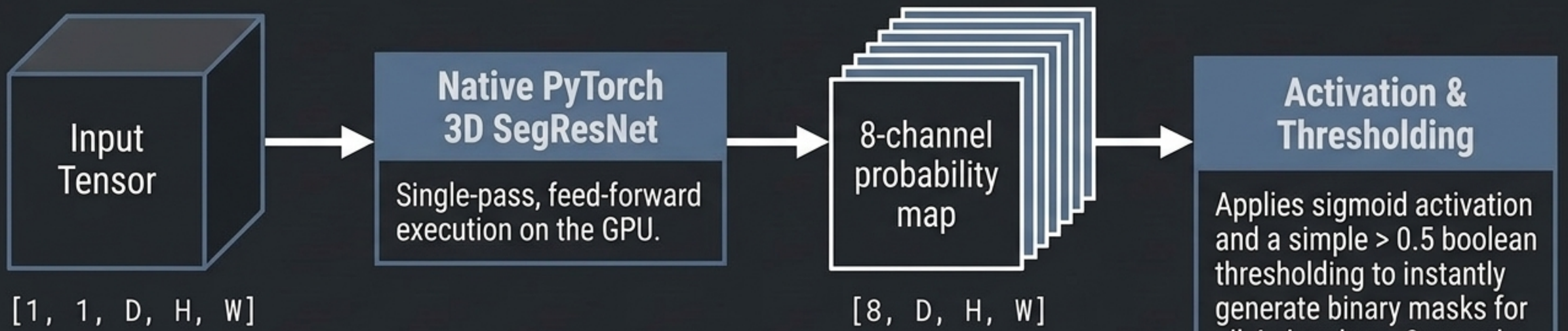
Step 2: The Orchestrator

The orchestrator dynamically checks user configurations, available `.onnx` or `.pt` models, and hardware capabilities to elegantly route the tensor to the optimal execution path.



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Path A: MONAI 3D SegResNet Backend

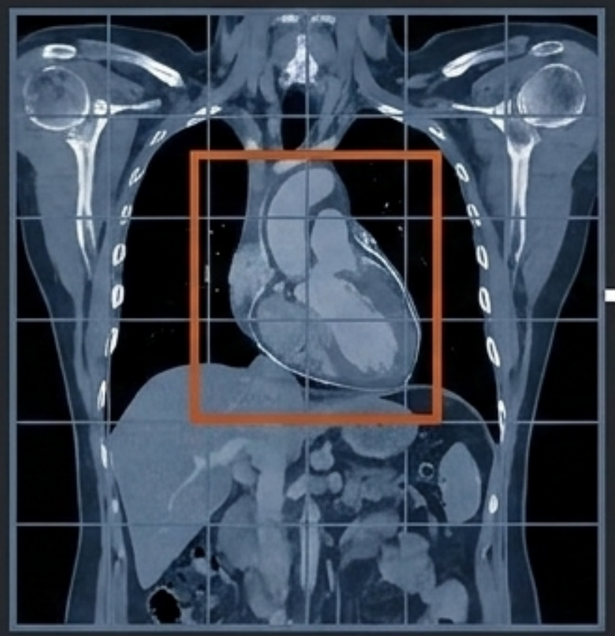


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Path B: ONNX 2-Stage Pipeline

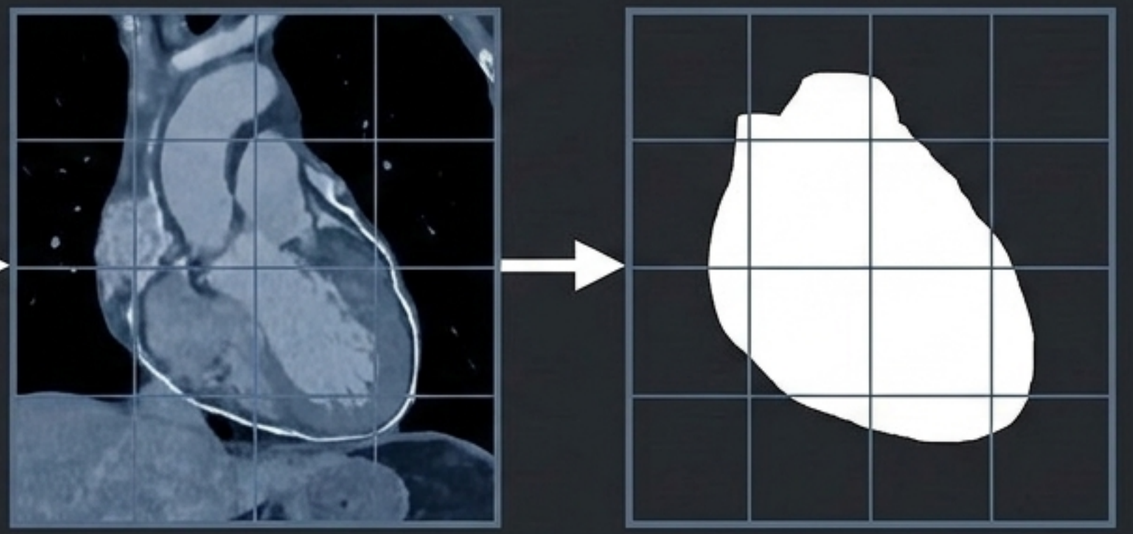
Required compatibility layer for legacy DeepHeartSeg single-channel frozen models.

Stage 1: Localization



Heavy downsampling (`ndi.zoom`) runs `deepheartseg_loc.onnx` to discover the cardiac bounding box in the full torso.

Stage 2: Fine Segmentation

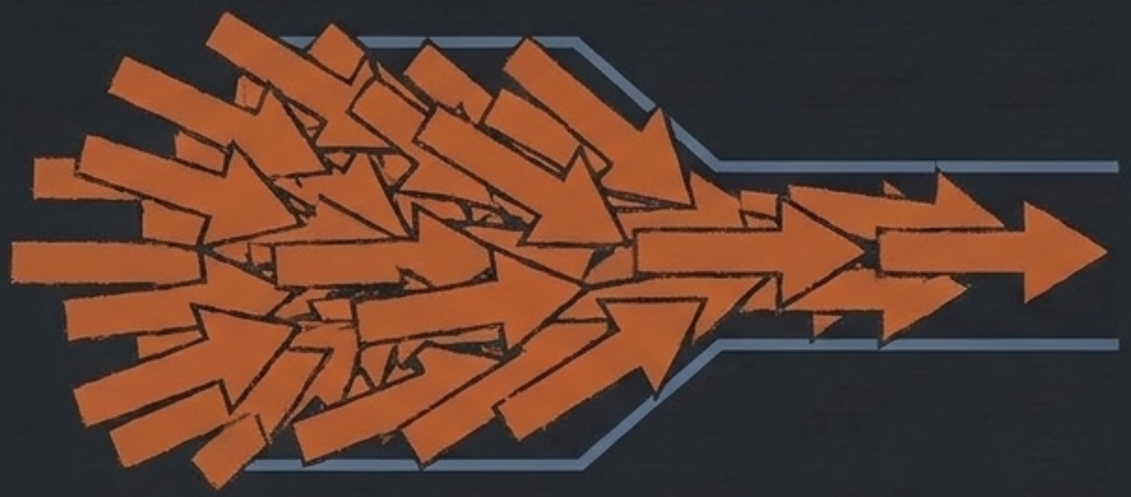


Crops original array with safety padding, resamples to exactly (112, 128, 128), runs `deepheartseg_heart.onnx`, and upsamples the binary pericardium mask back to native patient space.

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Optimizing Path B: ONNX Thread Control

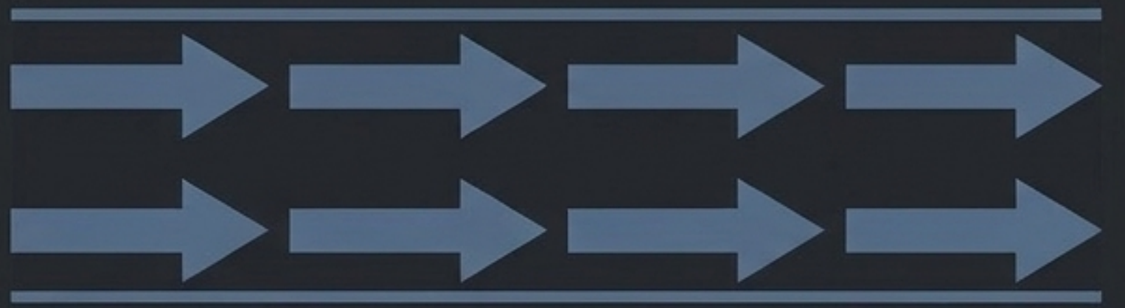
Thread Contention



Default ONNX settings on multicore nodes oversubscribe CPU threads, causing L3 cache thrashing, out-of-memory crashes, and OS freezing.

`128 competing threads`
(8 workers × 16 threads).

Explicit Control



The pipeline injects `ort.SessionOptions()` to strictly limit execution per worker:

```
intra_op_num_threads = 2
inter_op_num_threads = 1
```

Result: Safe multi-process throughput without host instability.

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Path C: The Zero-Download Heuristic Fallback

If model weights fail to download or an air-gapped node lacks internet access, the pipeline refuses to crash.

Raw CT Slice

↓

Step 1: Thresholding
Captures the mediastinum by mathematically isolating tissues between **-50** and **500 HU.**

↓

Step 2: Morphology
Applies `ndi.binary_closing` (2 iterations) to enclose chambers and the pericardial space, followed by hole filling.

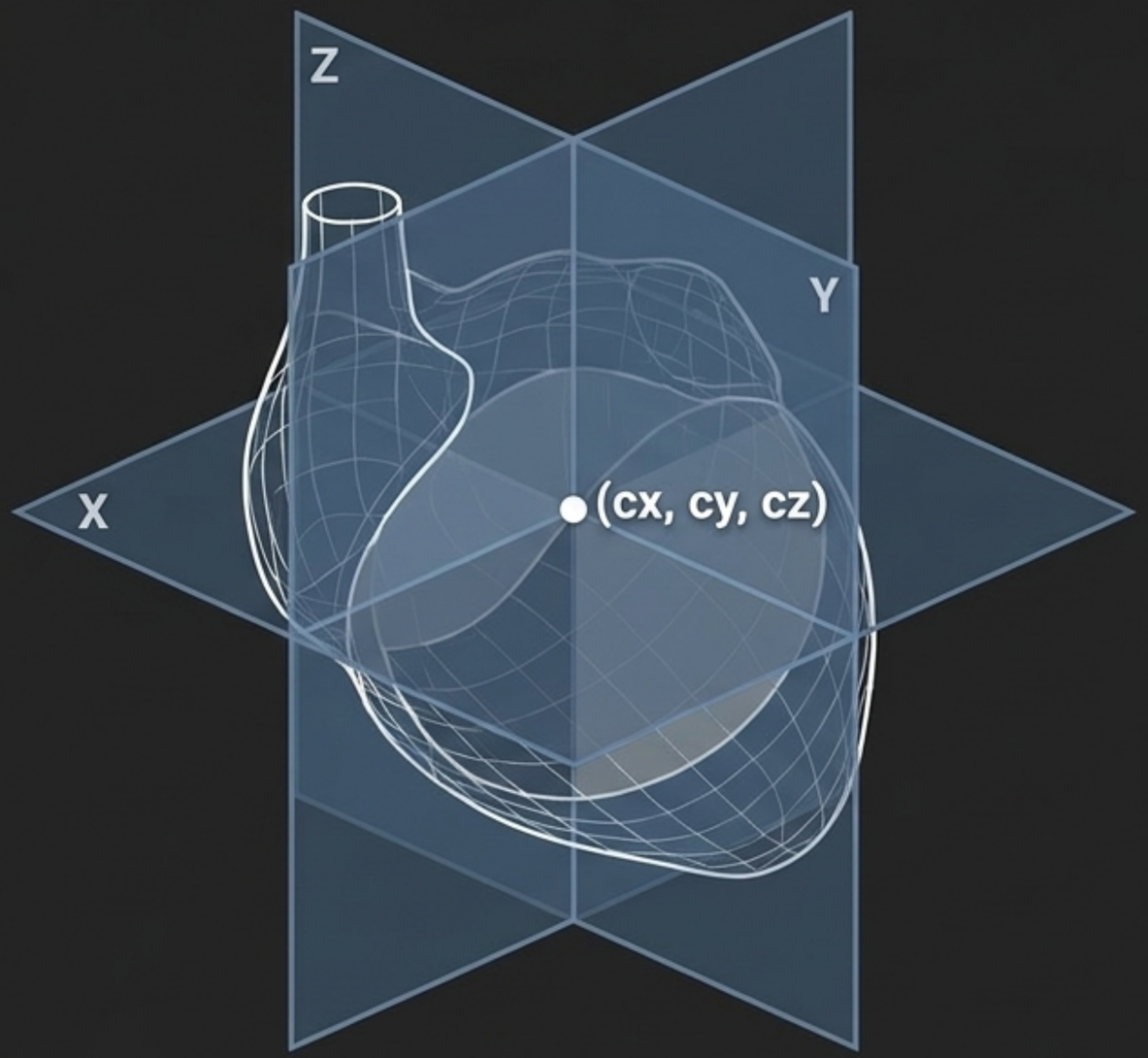
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Step 3: Component Isolation
Executes **3D connected components** to isolate the largest central mass, yielding the pericardial envelope purely via heuristics.

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Algorithmic Derivation: Spatial Quadrant Chamber Rules

How the pipeline derives 4 chambers from a 1-channel pericardium output using _derive_chambers_and_vessels.



The algorithm calculates the 3D centroid and applies boolean array slicing:

LA (Left Atrium)

```
is_left & is_posterior & is_superior
```

LV (Left Ventricle)

```
is_left & (is_inferior
        | (is_posterior & ~is_superior))
```

RA (Right Atrium)

```
is_right & (is_superior
         | (is_anterior & is_superior))
```

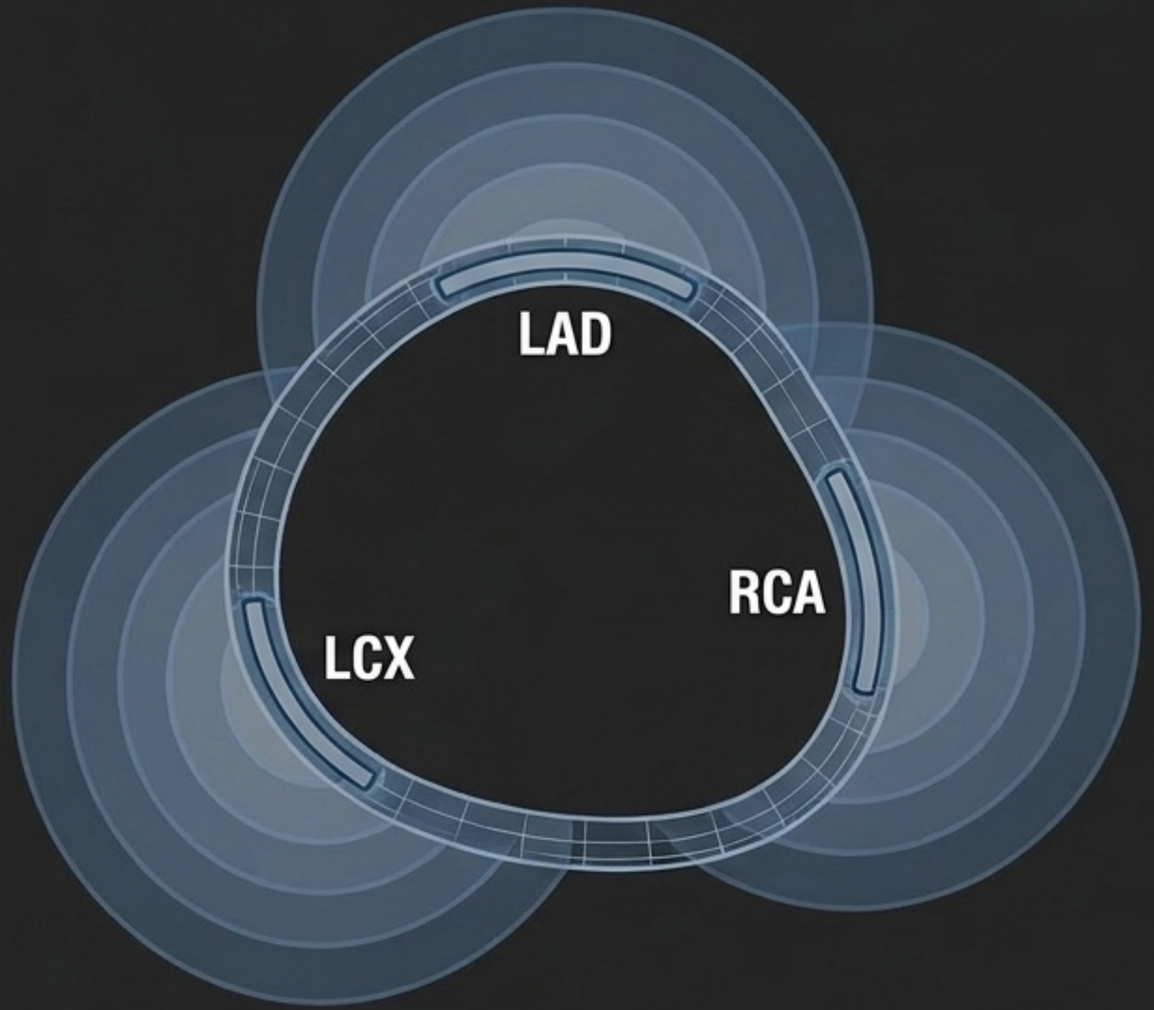
RV (Right Ventricle)

```
is_right & is_anterior & is_inferior
```

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Algorithmic Derivation: Coronary Trajectories

Deriving pericoronary fat (PVAT) boundaries via geometric grooves and radial dilation.



Defining the Spatial Zones:

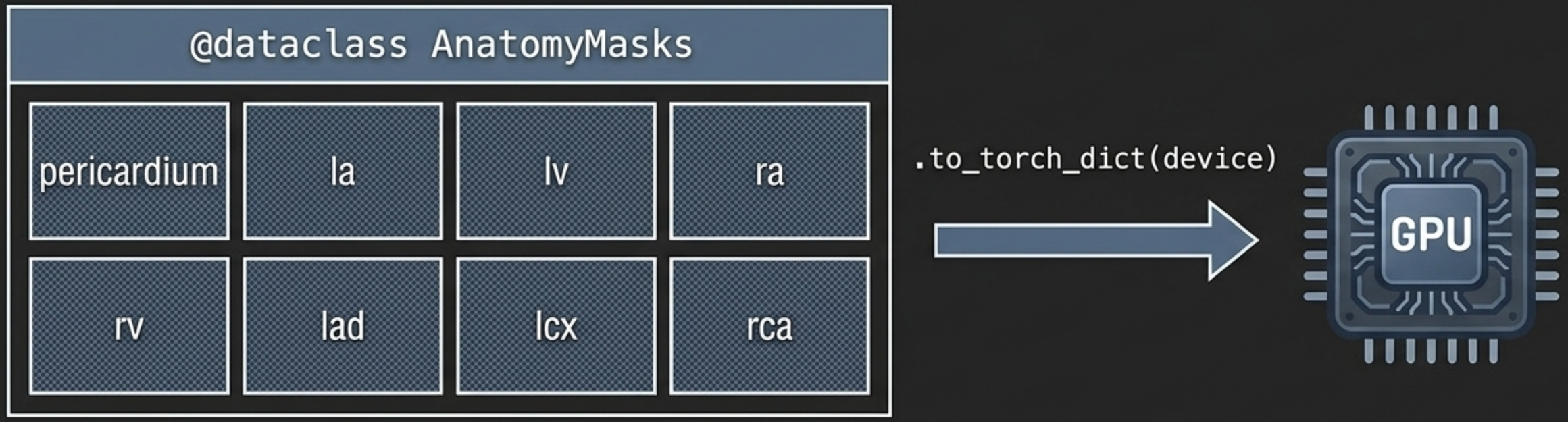
- **LAD**: Anterior margin within 8 voxels of the X-centroid.
- **LCX**: Left posterior margin within 8 voxels of the Z-centroid.
- **RCA**: Right anterior margin within 8 voxels of the Z-centroid.

The Derivation Process:

- Applies `ndi.binary_dilation` (expanding 1 iteration outward).
- Intersects with a **boolean AND** mask where `HU >= -30.0` to strictly isolate the vascular structures within the geometric groove.

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Step 3: The Output Container

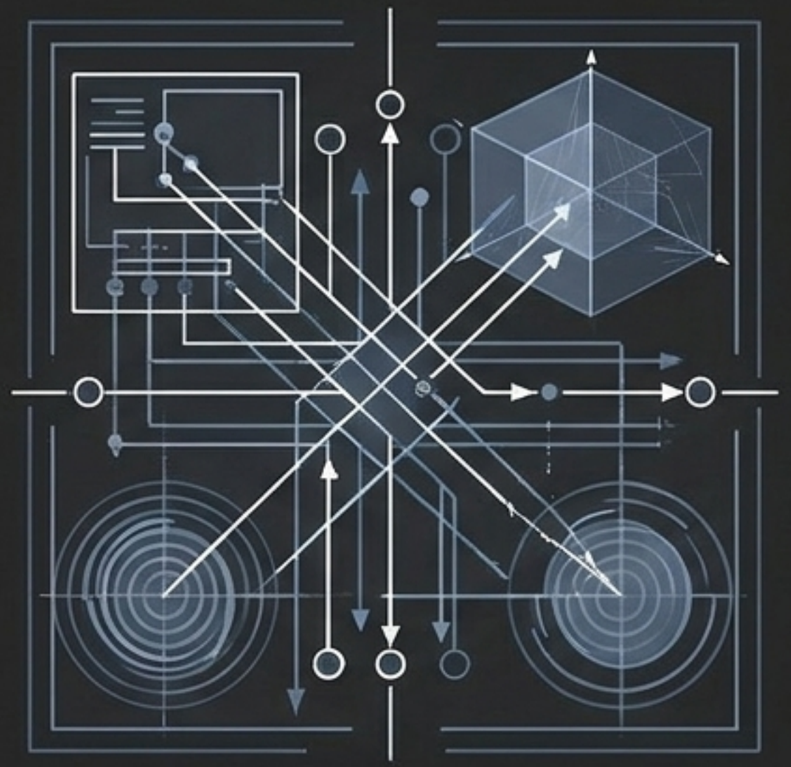


The execution flow terminates at this unified terminal, standardizing outputs from all three backends.

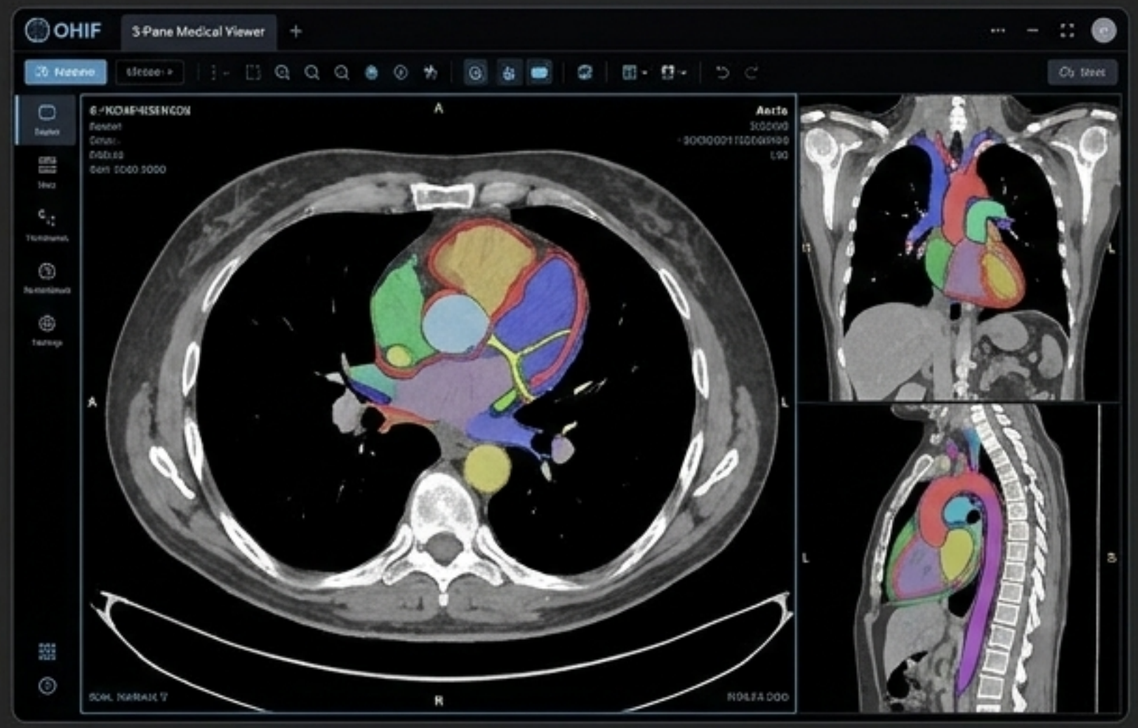
Instantly marshals CPU numpy arrays into GPU PyTorch tensors, preparing them for NVFlare statistical aggregation and accelerated distance transforms.

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Summary: Architecture Enables Clinical Scale



Encapsulated Complexity: Dynamic routing, thread safety, and geometric derivation.



Tangible Output: Seamless clinical application.

The Result: The overarching Federated Learning system achieves massive clinical throughput (`120+ scans/min on an RTX 4090`) while remaining 100.0% mathematically equivalent to the legacy baseline. Invisible, scalable, and clinically exact.

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